HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Managed production quality support for electromechanical hardware systems, identifying quality trends through **data analysis** and **data mining** on ~5,000 production units per quarter.
- Developed and maintained **KPIs** and **OKRs** for product quality tracking, presenting findings to a cross-functional team to drive targeted improvements.
- Led a project to implement a new inspection method for electromechanical components, reducing defect rates by **18%** through enhanced measurement protocols.
- Authored comprehensive **work instructions** and **quality plans** for hardware at scale, ensuring adherence to stringent manufacturing and assembly standards.
- Collaborated hands-on with technicians and engineers in a lab environment, utilizing **Power Supplies**, **Electronic Loads**, and **Digital Volt Meters** to build and measure prototype hardware.
- Supported the quality management system by managing changes to **documentation** and **training materials** for production line staff, ensuring consistent process application.
- Wrote **Python scripts** to automate data collection and analysis from test fixtures, improving data processing efficiency by approximately **30%**.

SKILLS

Quality Engineering – Quality Management System, KPI Development, OKR Tracking, Process Improvement, Documentation Control, Training Materials

Data Analysis & Mining – Data Analysis, Data Mining, Trend Identification, Pattern Recognition, Statistical Analysis

Hardware & Lab Work – Electromechanical Hardware, Prototype Building, Measurement, Power Supplies, Electronic Loads, Digital Volt Meters

Software & Scripting – Python, Shell Scripting, Data Collection Automation

Project Management – Project Leadership, Cross-functional Team Collaboration, Change Management

PROJECTS

Automated Quality Data Pipeline for Defense Systems

- Engineered a Python-based data pipeline to ingest and process quality test data from ~2,000 electromechanical units monthly, replacing manual data entry.
- Developed automated reports and dashboards using **Power BI** to visualize key quality metrics, enabling faster identification of system anomalies for engineering review.
- Integrated the pipeline with existing Jira ticketing for defect tracking, reducing the average time to log a new issue by **22%**.

Hardware Production Issue Triage and Resolution

- Supported production issue resolution for a complex hardware system, performing root cause analysis on ~15 critical production defects over a 4-month period.
- Developed standardized troubleshooting guides and implemented process improvements that decreased P1 incident resolution time by an average of **1.5 hours**.
- Collaborated with cross-functional teams including manufacturing and design engineering to implement corrective actions and validate fixes.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)